

Introduction to AI&ML (Open Elective) – 21AM8X95

Question Bank

UNIT I

1. Define Artificial Intelligence. Explain the different approaches for Artificial Intelligence in detail.
2. Briefly explain the different branches of Artificial Intelligence.
3. Describe the characteristics of Expert system and write the limitations
4. Write short note on a) Fuzzy logic b) Natural Language Processing
5. Define Task environment and explain its properties
6. What is task environment? How it is specified?
7. Explain the toy problems by considering example of vacuum world and 8 Queens 'Problem
8. Apply "solving problem by searching" strategy to solve 8-Queen's problem.
9. Explain the various applications of artificial Intelligence (State of the art).
10. Explain Simple reflex agent with suitable diagram.
11. Explain Goal Based Agent in detail
12. Explain Model Based Reflex Agent.
13. Explain Utility Based Agent in detail.

MCQs

1. Which of the following best defines Artificial Intelligence?

A) Programming computers to make them faster

B) Making computers mimic human intelligence

C) Designing chips for better computation

D) Writing algorithms for databases

2. Which AI approach is based on laws of thought?

A) Acting Humanly

B) Thinking Humanly

C) Thinking Rationally

D) Acting Rationally

3. Which branch of AI deals with understanding and processing human language?

A) Robotics

B) Natural Language Processing

C) Expert Systems

D) Fuzzy Logic

4. An agent in AI is defined as:

A) Human interacting with machine

B) Anything that perceives the environment and acts upon it

C) Database system

D) Computer hardware

5. Which of the following is NOT a type of agent program?

A) Simple reflex

B) Model-based reflex

C) Learning agent

D) Compiler agent

6. A problem-solving agent is one that:

A) Uses random moves

B) Works without search

C) Finds a sequence of actions to achieve a goal

D) Avoids goal states

7. Airline route-finding is an example of:

A) Real-world problem

B) Toy problem

C) Constraint satisfaction

D) Reflex agent

8. In AI, a well-defined problem requires:

A) Initial state

B) Set of actions

C) Goal test

D) All of the above

9. The primary goal of Artificial Intelligence (AI) is:

A) To make computers run faster

B) To enable machines to simulate human intelligence

C) To design more efficient hardware

D) To create database algorithms

10. Which of the following is NOT a domain of AI?

A) Computer Vision

B) Robotics

C) Natural Language Processing

D) Statistical Data

11. What does the Turing Test aim to evaluate?

A) The efficiency of an AI system

B) The ability of an AI system to act rationally

C) Whether a machine can exhibit intelligent behavior equivalent to, or indistinguishable from, a human

D) The programming language used in AI systems

12. A simple reflex agent acts based on:

A) Learning

B) History of states

C) Current percept only

D) Prediction

13. The Vacuum World problem is primarily used to illustrate:

A) Reflex agents

B) Complex decision-making

C) Problem-solving using search

D) Learning agents